Physics Worksheet

Derivative Practice

Name: __________________ **Student #:** ______________ **Student Email #:** _______

1. Compute: $$\frac{d}{dx}(x^5 + 3x^3 + \frac{1}{x})$$

2. Compute: $$\frac{d}{dx}(\frac{\cos x}{\sin x})$$

3. Compute: $$\frac{d}{dx}\ \ln(\cos(x))$$

4. Compute $\frac{d}{dx}x^3$ using the limit definition of the derivative

5. You are standing on a mountain who's shape can be modelled as $z = 7 - \frac{1}{3}x^2 - 2y^4$. You are at the point of $(2, 1, \frac{13}{3})$ and wish to descend the mountain as quikcly as possible. What direction (in unit vector form) should you go in?

6. Below is a plot of the function $y = x^2$, compute and plot on the following two empty graphs the derrivatives: $\frac{d}{dx}(x^2)$ and $\frac{d^2}{dx^2}(x^2)$